PNAS RESEARCH ARTICLE | PHYSIOLOGY
PSYCHOLOGICAL AND COGNITIVE SCIENCES

Sleep duration and timing are associated with next-day physical activity: Insights from two large-scale wearable sensor studies

Josh Leota[a,b,1], Mark É. Czeisler[c], Flora Le[a], David M. Presby[d,e,f], Emily R. Capodilupo[d], Hannah Scott[g], Joshua F. Wiley[a], Sean P. A. Drummond[a], Shantha M. W. Rajaratnam[a], and Elise R. Facer-Childs[a,1]

Affiliations are included on p. 9.

Edited by Emmanuel Mignot, Stanford University School of Medicine, Palo Alto, CA; received October 10, 2024; accepted May 12, 2025

Sleep and physical activity (PA) are pillars of health. However, the temporal dynamics between these two behaviors remain poorly understood. This research aims to examine the independent and interactive between- and within-person associations of sleep duration and sleep onset timing on next-day PA duration in two large, longitudinal samples of adults under free-living conditions. In the primary study, participants (N = 19,963; 5,995,080 person-nights) wore a validated biometric device (WHOOP) for 1 y (01/09/2021 to 31/08/2022). Objective sleep and PA metrics were derived from the wrist-worn device. Generalized additive mixed models assessed between- and within-person associations between sleep and PA variables, adjusted for age, sex, Body Mass Index, weekday/weekend, seasonal effects, biometric feedback, and autocorrelated errors. Between participants, longer sleep duration and later sleep onset timing were associated with decreased moderate-to-vigorous PA (MVPA) and overall PA duration (*ps* < 0.001). Within participants, sleeping shorter-than-usual and falling asleep earlier-than-usual were associated with increased next-day MVPA and overall PA, whereas sleeping longer-than-usual, or falling asleep later-than-usual, showed the opposite relationship (*ps* < 0.001). Next-day MVPA duration was highest following earlier-than-usual sleep onset timing *combined* with one's typical sleep duration. Results were consistent but smaller in magnitude in the external validation study (N = 5,898; 635,477 person-nights) using Fitbit data from the All of Us Research Program. Individuals may sacrifice time in one health behavior for time in the other. Interventions promoting exercise and holistic public health messaging should consider the temporal dynamics between sleep and next-day PA outcomes.

sleep health | exercise | health behaviors | exercise participation

Significance

Physical activity (PA) has been widely recommended to improve health. Uncovering upstream determinants of PA can help identify useful, modifiable targets to increase PA participation. We investigated whether sleep duration and sleep timing were associated with the duration of moderate-to-vigorous PA and overall PA the following day. Results from two independent longitudinal studies showed that both average sleep and nightly changes in sleep are related to PA. Going to sleep earlier than usual may be an effective strategy to maintain normal sleep duration while also optimizing next-day PA. Public health messaging should consider the dynamic interplay of sleep and PA by developing comprehensive and holistic recommendations that promote both health behaviors in ways that are effective and mutually beneficial.

Sufficient sleep and regular physical activity (PA) are foundational to a healthy life (1, 2). Both are individually recommended as modifiable behaviors for treating chronic diseases and promoting physical and psychological health (3, 4). Together, they are thought to share a complex, synergistic relationship where one behavior builds on and reinforces the other, triggering better—or worse—health outcomes. For example, PA promotes sleep through the regulation of circadian rhythms and cardiac and autonomic functioning (5, 6), while physical inactivity is associated with elevated risk of insomnia and sleep disturbance (7). Sleep loss is associated with reduced athletic performance (8), increased injury risk (9), and impaired physiological recovery from strenuous PA (10). Understanding the temporal dynamics of sleep and PA is critical for developing comprehensive and holistic public health messaging. This importance is further underscored by the fact that one in three adults fails to achieve the recommended 7 h of sleep per night (11), and four in five adults are insufficiently active (i.e., fail to engage in at least 150 to 300 min of moderate-intensity aerobic activity or 75 to 150 min of vigorous-intensity activity per week) (12).

Prior experimental and epidemiological studies on the temporal relationship between sleep and PA have shown inconsistent results (13). Experimental studies showed induced sleep restriction lowered next-day PA (14, 15). Conversely, some epidemiological studies showed that shorter sleep duration was associated with more PA (16–19), whereas others have reported no relationship (20–22). Sleep loss and circadian misalignment can disrupt mood and self-regulation, which may inhibit motivation to engage in PA (23–25). On the other hand, sleeping longer may reduce the time available during the waking day for PA (26).

Little is known about the associations between sleep and subsequent moderate-to-vigorous PA (MVPA) in the naturalistic setting of everyday life. Controlled experimental studies involve *prescribed* next-day MVPA sessions and typically focus on performance and

Competing interest statement: D.M.P. and E.R.C. are affiliated with the commercial company WHOOP, Inc. which provided support in the form of salaries. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as potential competing interests.

This article is a PNAS Direct Submission.

Copyright © 2025 the Author(s). Published by PNAS. This article is distributed under Creative Commons Attribution-NonCommercial-NoDerivatives License 4.0 (CC BY-NC-ND).

[1]To whom correspondence may be addressed. Email: josh.leota@monash.edu or elise.facer-childs@monash.edu.

This article contains supporting information online at https://www.pnas.org/lookup/suppl/doi:10.1073/pnas.2420846122/-/DCSupplemental.

Published June 30, 2025.

physiological consequences of sleep loss. Similarly, athlete sleep studies also involve prescribed MVPA (e.g., next-day training or competition). In both scenarios, participants are denied the opportunity to avoid MVPA entirely and therefore these studies do not reflect free-living conditions. These studies also do not consider the between- (inter-individual differences) and within-person effects (intra-individual differences, i.e., how an individual deviates from their habitual level) of sleep on MVPA. Conceptual models suggest these two sources of variability provide unique information about longitudinal processes (27).

The epidemiological literature is further clouded by differences in methodological and statistical approaches. Most studies use self-reported measures of sleep and MVPA which are subject to recall biases (13). Studies also ignore the potential impact of the outcome variable autocorrelation (13, 17, 22). For example, whether an individual engages in MVPA today may influence their likelihood of engaging in MVPA tomorrow, independent of their sleep. It is therefore important to account for such autocorrelation to provide a more robust test of the directionality between sleep and MVPA. Most studies involve short monitoring periods of 4 to 14 d, making it challenging to account for weekday and seasonal effects on sleep and MVPA (28, 29). Critically, there is a reliance on linear modeling despite evidence of the nonlinear relationship between sleep and health showing both short and long sleepers are at increased risk of adverse health outcomes (30–32). Finally, no study to our knowledge has investigated the daily relationship between sleep onset timing and MVPA in adults.

We address these gaps using objectively measured data in two large, longitudinal, free-living samples of adults to examine the between- and within-person associations of sleep duration and sleep onset timing with next-day MVPA and overall PA. The primary study included physically active adults who subscribed to the biometric device platform WHOOP, Inc. (Boston, MA). To evaluate the generalizability of the primary study's findings, an external validation study was performed on a cohort of adults from the All of Us Research Program.

Results

Primary Study (WHOOP) Results.

Demographics and descriptive statistics. The study population included 19,963 participants logging an average of 323.91 d of data (SD = 38.80) across the 1-y study. Baseline sample characteristics are presented in Table 1. A summary of the model parameters for MVPA duration (*SI Appendix*, Table S1) and overall PA duration (*SI Appendix*, Table S2) are provided in the *SI Appendix*.

Sleep Duration.

Between-person. There were significant negative associations between average sleep duration and MVPA and overall PA durations (Fig. 1 *A* and *B*; see *SI Appendix*, Table S5 *A* and *B* for contrasts and *SI Appendix*, Table S3 *A* and *B* for estimated marginal means). Participants with an average sleep duration of 7 h (reference level) had an 86.72-min average duration of daily MVPA and a 374.28-min average duration of overall PA. Participants who slept shorter than 7 h on average engaged in 10.39 to 31.18 more minutes of MVPA and 17.40 to 39.15 more minutes of overall PA compared to the reference level ($ps < 0.001$). Participants who slept longer than 7 h on average engaged in 10.39 to 31.16 fewer minutes of MVPA and 23.52 to 74.37 fewer minutes of overall PA compared to the reference level ($ps < 0.001$). Compared to participants who slept 9 h on average, participants who slept 5 h on average engaged in 41.56 more minutes of MVPA and 78.19 more minutes of overall PA ($ps < 0.001$).

Table 1. Primary study (WHOOP) participant characteristics

Variable	Mean ± SD	No. (%)
Total		19,963 (100)
Gender		
Women		5,203 (26.06)
Men		14,760 (73.94)
Age, y	38.44 ± 10.74	
18–44		14,475 (72.51)
44–64		5,129 (25.69)
64–87		359 (1.80)
Weight, KG	81.41 ± 16.23	
Women, KG	67.88 ± 13.26	
Men, KG	86.17 ± 14.43	
Height, CM	176.39 ± 9.42	
Women, CM	165.82 ± 7.06	
Men, CM	180.12 ± 7.02	
BMI (KG/M^2)	26.09 ± 4.19	
Women, (KG/M^2)	24.76 ± 4.56	
Men, (KG/M^2)	26.55 ± 3.95	
Sleep variables		
Sleep duration (h)	6.93 ± 0.63	
Sleep onset timing (hh:mm)	23.16 ± 1.18	
PA variables		
Days of data	323.91 ± 38.80	
MVPA duration	90.58 ± 52.45	
Overall PA duration	365.81 ± 103.88	

Note. SD: standard deviation; No.: number; y: years; KG: kilogram; CM: centimeter; h: hours.

Within-person. There were significant nonlinear associations between nightly sleep duration changes and next-day MVPA and overall PA durations (Fig. 1 *C* and *D*; see *SI Appendix*, Table S5 *C* and *D* for contrasts and *SI Appendix*, Table S3 *C* and *D* for estimated marginal means). Following nights when participants obtained a sleep duration unchanged from their monthly mean (i.e., 0-h change; reference level), they had an 87.40-min duration of next-day MVPA and a 375.66-min duration of next-day overall PA. Following nights when participants obtained below-average sleep durations, MVPA was 5.94 to 9.66 min higher and overall PA was 20.45 to 35.50 min higher compared to the reference level. Conversely, following nights when participants obtained longer-than-average sleep durations, MVPA was 6.41 to 28.00 min lower and overall PA was 20.42 to 93.21 min lower compared to the reference level. The duration of next-day MVPA was longest (9.66-min increase) following a 2-h shorter-than-average sleep duration and shortest (28.00-min decrease) following a 4-h longer-than-average sleep duration.

Sleep Onset Timing.

Between-person. There was a significant negative linear association between later average time of sleep onset and average daily MVPA duration but no association with average daily overall PA duration (Fig. 2 *A* and *B*; see *SI Appendix*, Table S6 *A* and *B* for contrasts and *SI Appendix*, Table S4 *A* and *B* for estimated marginal means). Participants with an average sleep onset timing of 11:00 PM (reference level) had an 88.55-min average duration of daily MVPA and a 375.88-min average duration of overall PA. Participants with an average sleep onset timing 1 to 3 h earlier than 11:00 PM engaged in 7.07 to 21.21 more minutes of MVPA ($ps < 0.001$). In contrast, participants with an average sleep onset timing 1 to 3 h later than 11:00 PM engaged in 7.07 to 21.21 fewer minutes of MVPA ($ps < 0.001$). Compared to participants with an average sleep onset timing of 1:00 AM, participants with

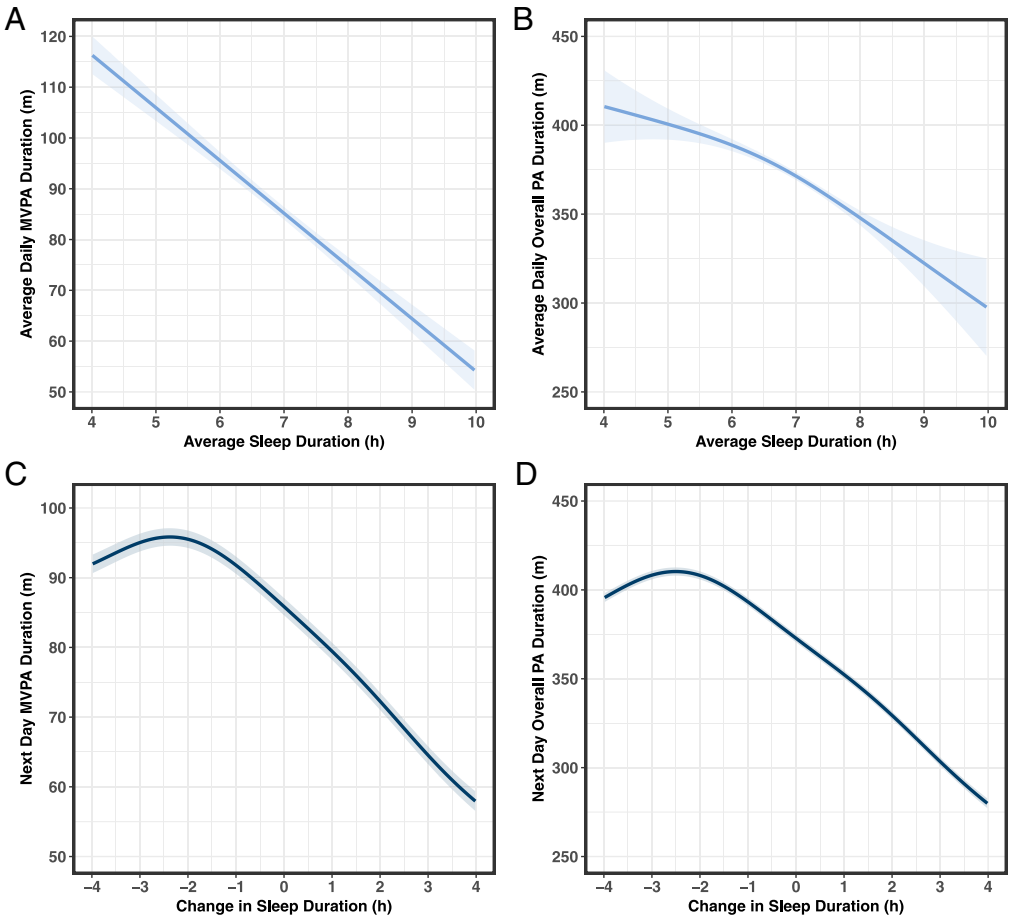

Fig. 1. GAMMs demonstrating the between-person relationships between average sleep duration and average daily (*A*) MVPA duration and (*B*) overall PA duration, and the within-person relationships between nightly change in sleep duration and next day (*C*) MVPA duration and (*D*) overall PA duration. The shaded area surrounding each line represents 95% CI. m: minutes; h: hours.

an average sleep onset timing of 9:00 PM engaged in 28.28 more minutes of MVPA (*ps* < 0.001).

Within-person. There were significant associations between nightly sleep onset timing changes and next-day MVPA and overall PA durations (Fig. 2 *C* and *D*; see *SI Appendix*, Table S6 *C* and *D* for contrasts and *SI Appendix*, Table S4 *C* and *D* for estimated marginal means). Following nights when participants had a sleep onset timing unchanged from their monthly mean (i.e., 0-h change; reference level), they had an 87.39-min duration of next-day MVPA and a 375.63-min duration of next-day overall PA. Following nights when participants fell to sleep earlier than usual, MVPA was 6.26 to 13.78 min higher and overall PA was 20.47 to 64.35 min higher compared to the reference level. Conversely, following nights when participants fell to sleep later than usual, MVPA was 7.77 to 30.07 min lower and overall PA was 23.50 to 98.34 min lower compared to the reference level.

Sleep duration by sleep onset timing interaction. Nightly sleep durations *close to* the participant's monthly average (±30 min) combined with nightly sleep onset timings 1.5 h to 3 h *earlier* than the participant's monthly average were associated with the highest next-day MVPA duration (Fig. 3*A*) and next-day overall PA duration (Fig. 3*B*).

External Validation Study (All of Us) Results.

Demographics and descriptive statistics. The study population included 5,898 participants with a median 108 d of recording across the 120 d study interval. Baseline sample characteristics are presented in Table 2. A summary of the model parameters for MVPA duration (*SI Appendix*, Table S7), overall PA duration (*SI Appendix*, Table S8), and step count (*SI Appendix*, Table S9) are provided in the *SI Appendix*.

Sleep Duration.

Between-person. There was a significant nonlinear association between average sleep duration and average MVPA duration (Fig. 4*A*) that was similar to the primary study (WHOOP). Participants with an average sleep duration of 7 h (reference level) had a 38.16-min duration of MVPA. Participants with an average sleep duration of 6 h or shorter had a similar average duration of MVPA compared to the reference level (*ps* ≥ 0.07). Participants who slept longer than 7 h on average had a 13.29 to 46.37% (5.06 to 17.79 min) shorter average duration of MVPA (*ps* ≤ 0.01). Compared to participants who slept 9 h on average, participants who slept 5 h on average had a 57.37% (14.79 min) longer average duration of MVPA (*P* < 0.001).

Within-person. There was a significant nonlinear association between nightly sleep duration changes and the duration of next-day MVPA (Fig. 4*C*) that was similar to the primary study (WHOOP). Following nights when participants obtained a sleep duration unchanged from their monthly mean (i.e., 0-h change; reference level), they had a 39.85-min duration of MVPA. The duration of next-day MVPA was unchanged following a 1-h to 4-h shorter-than-average sleep duration. The duration of next-day MVPA was persistently lower (– 2.74 to –14.75%; –1.09 to –5.88 min) when participants slept 1 to 4 h longer than their average (*ps* < 0.001).

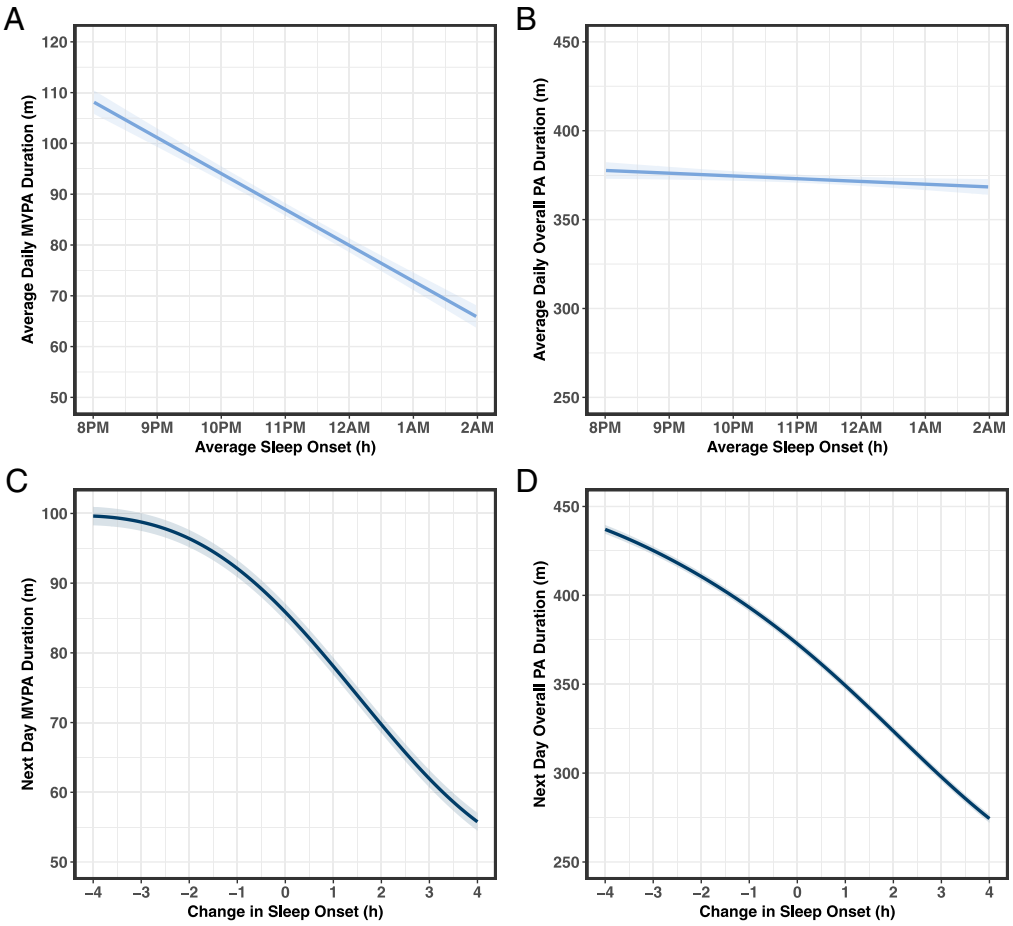

Fig. 2. GAMMs demonstrating the between-person relationships between average sleep onset and average daily (*A*) MVPA duration and (*B*) overall PA duration, and the within-person relationships between nightly change in sleep onset and next day (*C*). MVPA duration and (*D*) overall PA duration. The shaded area surrounding each line represents 95% CI. m: minutes; h: hours.

Sleep Onset Timing.

Between-person. There was a significant, nonlinear association between average sleep onset timing and average MVPA duration (Fig. 4*B*). Participants with an average sleep onset timing of 11:00 PM (reference level) had a 43.09-min duration of MVPA. Participants with an average sleep onset timing 1 to 3 h earlier than 11:00 PM had a 11.46 to 39.80% (4.93 to 17.15 min) longer average duration of MVPA (*ps* < 0.001). In contrast, participants with an average sleep onset timing 1 to 3 h later than 11:00 PM had an 8.54 to 14.38% (−3.68 to −6.20 min) shorter average duration of MVPA (*ps* < 0.001). Compared to participants with an average sleep onset timing of 1:00 AM, participants with an average sleep onset timing of 9:00 PM had a 44.11% (16.48 min) longer average duration of MVPA (*ps* < 0.001).

Within-person. There was a significant nonlinear association between nightly sleep onset timing changes and next-day MVPA (Fig. 4*D*). Following nights when participants had a sleep onset timing unchanged from their monthly mean (i.e., 0-h change; reference level), they had a 39.82-min duration of MVPA. Earlier-than-average sleep onset timings were associated with 2.64 to 8.26% (1.05 to 3.29 min) longer durations of next-day MVPA (*ps* < 0.001), whereas later-than-average sleep onset timings were associated with 2.40 to 6.40% (−0.96 to −2. 41 min) shorter durations of next-day MVPA (*ps* < 0.001).

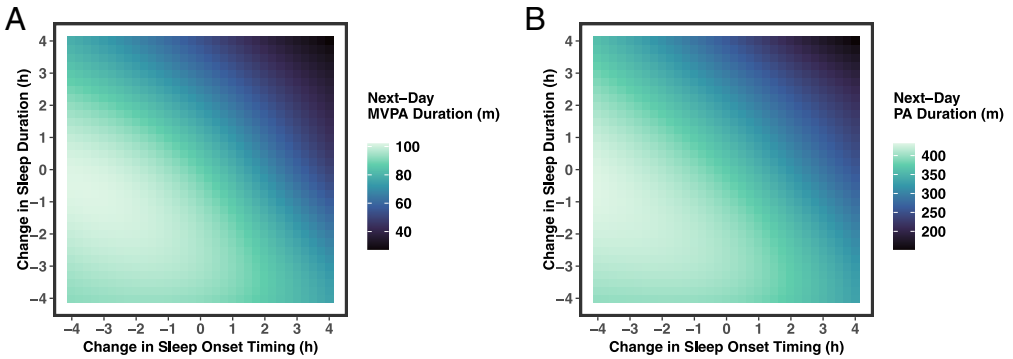

Fig. 3. Tensor product GAMMs demonstrating the within-person sleep duration by sleep onset timing interaction on (*A*) next-day MVPA duration and (*B*) next-day overall PA duration. m: minutes; h: hours.

Table 2. External validation study (All of Us) participant characteristics

Variable	Mean ± SD	No. (%)
Total		5,898 (100)
Sex		
Female		4,078 (69.14)
Male		1,821 (30.87)
Age, y–Median	59.0	
18–44		1,608 (27.26)
44–64		1,914 (32.45)
65–96		2,376 (40.28)
Combined race and ethnicity		
White, non-Hispanic		3,756 (63.68)
Black, non-Hispanic		524 (8.88)
Asian, non-Hispanic		480 (8.14)
Other race or multiple races, non-Hispanic		735 (12.46)
Hispanic or Latino, any race or races		403 (6.83)
Employment		
Employed		2,846 (48.25)
Unemployed		1,141 (19.35)
Retired		1,885 (31.96)
Annual income (USD)		
$50,000 or less		2,253 (38.2)
$50,000 to $150,000		2,592 (43.95)
$150,000 or more		763 (12.94)
Deprivation index	0.31 ± 0.05	
Cigarette smoking frequency		
<100 lifetime cigarettes		3,671 (62.24)
≥100 lifetime cigarettes, currently less than daily		1,878 (31.84)
≥100 lifetime cigarettes, currently daily usage		334 (5.66)
Alcohol use frequency		
Never or rarely		1,298 (22.01)
Weekly or less		3,083 (52.27)
Multiple weekly		1,500 (25.43)
Body Mass Index		
Underweight		75 (1.27)
Normal		1,748 (29.64)
Overweight		1,827 (30.98)
Obesity		2,248 (38.11)
Self-rated overall health		
Poor		218 (3.7)
Fair		1,040 (17.63)
Good		1,987 (33.69)
Very good		2,040 (34.59)
Excellent		602 (10.21)
Study characteristics		
Study interval start date–Median (IQR)	11/17/22 (11/22/21 to 05/09/23)	
Days of recording–Median (IQR)	108 (101 to 115)	
Sleep characteristics		
Sleep duration (h)	6.44 ± 1.03	
Sleep onset timing (hh:mm)	23.88 ± 1.66	
PA characteristics		
Moderate to vigorous PA duration (m)	34.16 ± 33.57	
Overall PA duration (m)	247.84 ± 88.01	
Step count	6,996.17 ± 3,826.55	

Note. SD: standard deviation; No.: number; y: years; h: hours; m: minutes; IQR: interquartile range; USD: United States Dollar.

Overall PA duration and step count. Between- and within-person associations between sleep duration and sleep onset and total PA duration (*SI Appendix*, Fig. S4 *A–D*) and step count (*SI Appendix*, Fig. S5 *A–D*) were consistent with MVPA duration. At the between-person level, shorter average sleep duration and earlier average sleep onset were associated with longer average overall PA duration and higher average step count. At the within-person level, sleeping shorter and earlier than usual was associated with increased next-day overall PA duration and step count, and sleeping longer and earlier than usual was associated with reduced next-day overall PA duration and step count.

Discussion

In the primary study analysis of objective sleep and PA data from ~6-million person-nights, we found that sleep duration and the timing of sleep onset were associated with MVPA and overall PA duration. At the between-person level, shorter sleep duration and earlier sleep onset timing were independently associated with increased MVPA duration. The highest-magnitude relationships were at the extremes. Individuals who typically slept 5 h compared to those who slept 9 h engaged in 41.5 more minutes of MVPA per day, while individuals who typically fell asleep at 9:00 PM compared to those falling asleep at 1:00 AM engaged in 28.3 more minutes of MVPA per day. At the within-person level, sleeping shorter-than-average duration and falling asleep earlier-than-average were independently associated with increased next-day MVPA (and overall PA) duration, whereas sleeping longer or falling asleep later than usual were associated with decreased next-day MVPA (and overall PA) duration. Finally, an earlier-than-average sleep onset timing combined with one's typical sleep duration was associated with the highest MVPA and overall PA durations the next day. An external validation study using participants from the All of Us Research Program showed similar associations across MVPA duration, overall PA duration, and step count. Together, our findings reveal robust relationships between sleep and PA patterns among adults in free-living conditions. The results are useful to inform evidence-based public health communication strategies about sleep and PA and to serve as a foundation for future studies designed to elucidate the mechanisms underlying the findings.

Sleep Duration. Sleep duration was differentially associated with MVPA at the between- and within-person levels. Empirical evidence in this area is currently divided. Some studies have failed to find between-person effects, reporting that this relationship may be solely dependent on within-person processes (16, 18). Conversely, several studies did not identify significant between- or within-person associations (13). By using nonlinear modeling methods, we provide evidence that this relationship may be linear or monotonic at between-person levels, but nonlinear at within-person levels. Therefore, the magnitude *and* direction of nightly sleep deviations from an individual's habitual level may be important considerations for predicting next-day MVPA behavior.

Our within-person findings extend prior work that showed 3 to 4 h of experimentally induced sleep restriction decreased PA under laboratory (14) and free-living conditions (15). We build on this work by showing that voluntarily reducing sleep by less than ~3 h (relative to one's average) was not associated with lower MVPA duration the next day in a naturalistic setting. Therefore, it is important to exercise caution when generalizing findings from sleep restriction studies to real-life situations involving smaller—and perhaps more tolerable—reductions in sleep duration. For example, people may sacrifice sleep to make time for exercise the

following day (e.g., waking up earlier to exercise before work), whereas greater sleep loss may reflect other lifestyle decisions (e.g., attending a social event at night), acute sleep difficulties, and/or begin to curtail one's motivation to exercise. When recommending exercise, physicians should consider the possibility that individuals may sacrifice sleep to make time for exercise. Our findings suggest that instead of shortening sleep, going to bed earlier may better position individuals to obtain next-day MVPA. Public health messaging aimed at increasing exercise adherence should prioritize recommendations compatible with healthy sleep behavior.

Consistent with prior research (17, 33, 34), sleeping longer than usual was strongly associated with decreased next-day MVPA (a 7.3 to 32.0% reduction) and overall PA (a 5.4 to 24.8% reduction) durations. Extending the sleep period will, by definition, reduce the available time for other behaviors, such as PA. Therefore, when other commitments are fixed and nonnegotiable (e.g., work, parental duties, etc.), planned PA may be abandoned or reduced in duration (35). Further, when navigating a demanding schedule, it is possible that chronically sleep-deprived individuals may sacrifice PA in favor of compensatory sleep. An increase in sleep duration may also reflect acute illness or vacation periods, which could also explain subsequent reductions in MVPA and overall PA. Conversely, prior research in athletes has shown experimentally prescribed sleep extension not only improved MVPA performance but also increased vigor and decreased fatigue and sleepiness (30), suggesting that sleep extension, under the right conditions (e.g., when there is sufficient time available for MVPA), may improve the duration of next-day MVPA. However, the unique schedules of athletes, where MVPA is prioritized, may not be realistic for the general population. Future research could explore the minimal effective dose of sleep extension interventions for next-day MVPA in an ecologically valid setting, given that more time asleep could come at the expense of time in waking activities (36).

Sleep Onset. Compared to the sample average of 11:00 PM, individuals with later average sleep onset timing demonstrated an 8.0 to 24.0% reduction in daily MVPA duration. Conversely, individuals with earlier average sleep onset timing demonstrated an 8.0 to 24.0% higher daily MVPA duration. One of the most striking findings of this study was that early sleepers with an average sleep onset timing of 9:00 PM engage in 38.0% more MVPA than late sleepers with an average sleep onset of 1:00 AM. These findings are consistent with prior research showing that late chronotypes or those with an eveningness preference are associated with less PA, more sedentary behavior (37), and poorer subjective fitness and exercise habits (38). Normal work schedules (e.g., 9:00 AM to 5:00 PM) can interfere considerably with the sleep preferences of evening chronotypes, resulting in social jetlag, impaired sleep quality, and daytime sleepiness (39, 40). For individuals with a late chronotype or eveningness preference, exercising before work may come at the cost of sleep duration and circadian alignment. Exercise at night, when late chronotypes ostensibly have more free time and energy, may also be suboptimal given late-night exercise, particularly involving higher cardiovascular strain, is associated with disruptions to subsequent sleep and autonomic function (41). To increase exercise participation, late chronotypes may benefit from restructuring their professional workday to facilitate opportunities for daytime MVPA, or by advancing their habitual sleep timing using a practical, nonpharmacological intervention (42).

Advancing nightly sleep onset timing by 1 to 3 h compared to one's average was associated with a 7.2 to 14.8% increased duration of next-day MVPA. Individuals may be sleeping earlier to compensate for early awakenings to facilitate morning MVPA, or simply exchanging waking hours today for waking hours tomorrow (and thus a greater MVPA opportunity window). Indeed, larger sleep onset advances were also associated with longer MVPA durations the next day. Critically, the days with the highest duration of MVPA were preceded by nights with earlier-than-average sleep onset timings *and* approximately normal sleep durations (Fig. 3). From a public health perspective, advancing sleep onset timing may provide a modifiable target for improving MVPA adherence and overall PA levels *without* compromising sleep duration.

Nightly sleep onset delays were negatively associated with next-day MVPA duration (−8.9 to −34.4%). Staying up later than usual is often associated with later wake up times and would therefore limit the available time to engage in MVPA the following day. This relationship may also be explained by external factors, such as late-night social commitments, and internal factors, such as acute illness or difficulties falling asleep. Two sleep restriction studies showed that PA (15) and step counts (43) decreased when sleep loss occurred at the beginning of the sleep period (i.e., delayed sleep onset timing) compared to the end (i.e., advanced sleep offset timing). We extend these findings by showing that delayed sleep onset under free-living conditions is associated with reduced MVPA levels even when adjusting for changes in sleep duration.

External Validation Study. To evaluate the generalizability of findings from the primary study (WHOOP), we conducted an external validation study using data from the *All of Us* Research Program; a large, longitudinal cohort study led by the NIH. This program is designed to enhance the representativeness of biomedical research by enrolling participants from historically underrepresented populations across the United States (44). A notable strength of the All of Us cohort, beyond its demographic diversity, is the availability of detailed information on demographic and lifestyle factors, enabling rigorous adjustment for potential confounders (Table 2). Results from the external validation study were broadly consistent with those of the primary study (WHOOP), albeit reduced in magnitude. This attenuation is likely attributable to lower average levels of MVPA in the All of Us cohort (34.16 min/d), compared to the WHOOP cohort (90.58 min/d).

Strengths and Limitations. This research has several strengths. First, the primary study (WHOOP) captured continuous, objective sleep and PA data for 365 d from almost 20,000 individuals, substantially exceeding the sample size and duration of most previous work. A major limitation of the current PA and sleep literature is the reliance on short sampling windows, often spanning only a few days (35), which limits the validity of daily models (17). By collecting longitudinal data across a full year, we were able to account for daily, weekly, and seasonal fluctuations in sleep–wake behaviors that might otherwise confound interpretation and limit generalizability. To further account for seasonal variation, we centered within-person sleep variables on participants' monthly means rather than their overall means, thereby capturing meaningful changes that covary with environmental factors such as photoperiod and temperature [e.g., longer nights and colder temperatures may promote sleep and reduce MVPA; (29)].

Second, we differentiated between- and within-person associations, enabling a rigorous test of directionality. This approach revealed that MVPA behavior was linked not only to stable, trait-like differences in sleep characteristics, but also to nightly fluctuations in an individual's sleep. These insights help clarify whether interventions should target acute changes in sleep behavior or focus on long-term modifications to habitual sleep patterns.

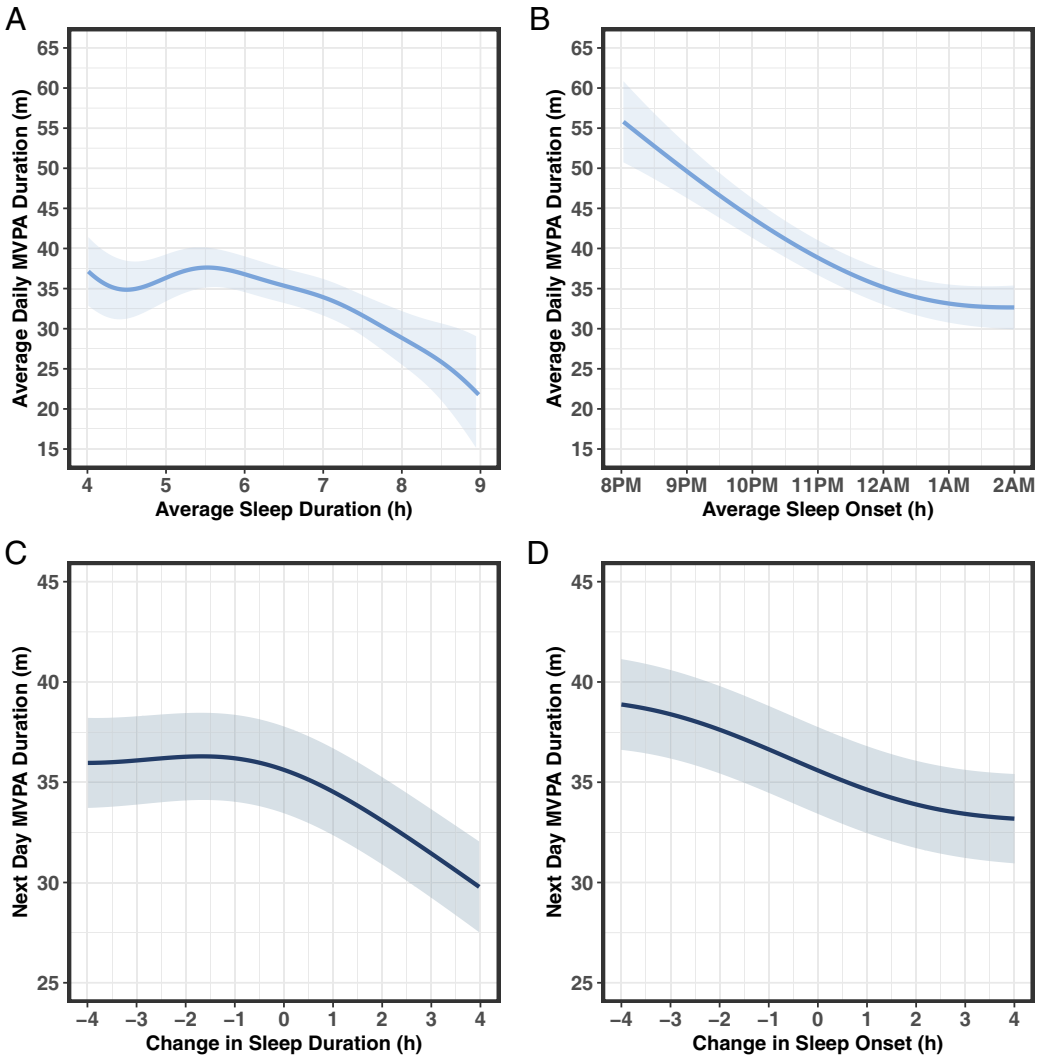

Fig. 4. GAMMs from the external validation study (All of Us) demonstrating the between-person relationships between (*A*) average sleep duration and average daily MVPA duration and (*B*) average sleep onset and average daily MVPA duration, and the within-person relationships between (*C*) nightly change in sleep duration and next-day MVPA duration, and (*D*) nightly changes in sleep onset and next-day MVPA. The shaded area surrounding each line represents 95% CI. m: minutes.

They also underscore the value of personalized intervention strategies that account for both how individuals differ from others and how they deviate from their own sleep norms on a given day.

Third, in two independent studies, we analyzed objective sleep and PA data using validated biometric devices. Objective and subjective sleep and PA data are weakly correlated (45), in part due to social desirability and recall biases (46), which might explain the inconsistent findings across studies (13).

This research also has limitations that should be addressed in future studies. First, the primary sample (WHOOP) may overrepresent individuals who are more physically active than the general population, potentially limiting generalizability. Given their subscription to a health and fitness wearable, it is also plausible that participants in the primary study prioritize MVPA. Further, limited demographic information precluded robust adjustments for potential confounds. However, the broadly consistent results observed in the external validation study, that included comprehensive adjustment for confounds, suggest that these associations may extend to other populations. Future studies should aim to replicate these findings in more diverse cohorts, including less active individuals, retirees, and clinical populations.

Second, the current studies did not assess certain variables worth investigating in future work. We did not assess long-term physical or mental health outcomes. Future longitudinal studies are required to investigate whether the relationships between sleep and MVPA have downstream consequences for broader health and well-being. We did not assess sleep quality, as this would require participants to subjectively report how they slept, or manually input when they first attempt sleep to calculate sleep efficiency. Future research should investigate the relationship between changes to sleep quality and next-day PA. Data for the primary study were collected in 2021 to 2022 and therefore may have been influenced by mobility restrictions relating to the COVID-19 pandemic; an important consideration when interpreting the results. We also did not have access to commute time information, which may encompass time that could otherwise be spent sleeping, given that leisure time MVPA often involves some number of incidental behaviors not captured by the device (e.g., travel to and from the gym). Additionally, although the WHOOP smartphone app generates a proprietary Recovery Score metric that may influence subsequent PA behavior, we were unable to ascertain whether participants viewed their Recovery Scores. To mitigate this, we conservatively included the Recovery Score as a covariate in our models, and we recommend that future studies using consumer wearable data adopt a similar approach.

Third, we did not examine associations between sleep and the timing of next-day PA, despite growing evidence linking morning

and evening PA to sleep and physical health outcomes (41, 47). Future work should explore this dimension, as well as the complex relationships between sleep, MVPA, and other daily behaviors, such as smartphone usage, which has been linked to subsequent sleep and PA previously (48, 49). Future studies could also examine whether regularity in sleep–wake patterns predicts consistency and volume of MVPA, given the link between sleep regularity and health (50). Finally, although this study describes the temporal relationship between sleep and MVPA using daily data, these data are still observational in nature and therefore cannot be used to infer causality. Further experimental research could build on our findings by examining whether sleep duration and timing interventions of different magnitudes effect change in next-day MVPA behavior.

Conclusion. We present evidence of between- and within-person associations between objectively measured sleep and next-day MVPA and overall PA from two large, independent samples of adults in free-living conditions. Between individuals, both longer average sleep duration and later average sleep onset timing were robustly associated with progressively lower average MVPA duration. Within individuals, the combination of going to sleep earlier than usual and getting approximately one's normal sleep duration was associated with the longest duration of next-day MVPA. Our findings have important implications for existing interventions and public health messaging. Interventions aimed at promoting MVPA should consider the role of sleep and, when necessary, opt for supporting earlier sleep onset timing to accommodate morning or daytime MVPA while maintaining sufficient sleep duration. Public health messaging should also consider the dynamic interplay of sleep and next-day MVPA by developing comprehensive and holistic recommendations that promote both health behaviors in ways that are effective and mutually beneficial.

Materials and Methods

Overview. The primary study cohort included 19,963 subscribers to WHOOP who provided continuous, timestamped objective sleep and PA data for 365 d resulting in nearly 6-million nights of data. To evaluate the generalizability of the primary study findings, particularly given limited demographic and health characterization of the WHOOP cohort, an external validation study was performed on a cohort of participants from the All of Us Research Program with objective sleep and PA data.

The primary study of WHOOP data was approved by Monash University Human Research Ethics Committee (#32928). The external validation study of All of Us data was approved after Harvard University Data Safety and Security Review (#DAT25-0107). The research adheres to the Strengthening the Reporting of Observational Studies in Epidemiology guidelines.

Primary Study (WHOOP) Methods.

Design, population, and data collection. Sleep and PA data over a 1-y interval (September 1, 2021, to August 31, 2022) were collected from 19,963 subscribers to WHOOP, Inc. (Boston, MA), a biometric device platform. Participants provided demographic information and informed consent for their deidentified data to be used for research purposes during registration with WHOOP. Participants were not specifically instructed of this analysis or associated hypotheses, minimizing the Hawthorne effect. For this analysis, participants were required to be ≥18 y of age and to have logged ≥328 d across the 1-y study (i.e., ≥90% usage). We further excluded days with <60% wear time. The final dataset included 5,995,080 person-nights of data (see CONSORT flow diagram, *SI Appendix*, Fig. S1*A*). Participant characteristics are presented in Table 1.

Sleep and PA metrics were derived from a wrist-worn multisensor device (WHOOP strap version 3.0 or version 4.0). The WHOOP biometric device is a wearable sleep and fitness tracker that uses photoplethysmography and a three-axis accelerometer to derive movement, heart rate, and oxygen saturation for sleep/wake categorization (51). The WHOOP device allows for uninterrupted physiologic monitoring with high temporal resolution and demonstrated 86 to 89% agreement with gold standard polysomnography for the assessment of sleep and wake for 2-stage (i.e., sleep versus wake) categorization (51, 52) and has been validated against electrocardiography for the assessment of heart rate (51). During onboarding, participants were instructed to wear the device continuously on their nondominant wrist, 1-inch superior to the head of the ulna.

Predictors: Sleep duration and sleep onset timing. Sleep duration and sleep onset timing were detected using the WHOOP Sleep Auto-Detection feature based on heart rate, heart rate variability, and activity patterns (53). Sleep duration was operationalized as the total time the participant spent asleep from sleep onset to sleep offset, minus any wake time during the sleep episode (i.e., total sleep time excluding day-time naps). Sleep onset timing was operationalized as the initial time of sleep. Between-person terms represented the participant grand sleep duration/onset timing mean. Within-person sleep terms were person-centered at the participant monthly mean (i.e., the mean sleep duration or onset timing of the calendar month of each night) as both sleep and MVPA can vary by season, though results are extremely similar when centering at the participant grand mean (*SI Appendix*, Figs. S2 and S3).

Outcomes: MVPA and overall PA duration. Continuous heart rate data derived from the WHOOP device were used to estimate MVPA and overall PA duration for the following day. MVPA was defined as the total time per day spent above 40% of an individual's heart rate reserve (HRR), while overall PA included all time spent above 20% HRR. These thresholds were based on thresholds published in ref. 54, where 20 to 39% HRR represents light effort, 40 to 59% HRR represents moderate effort, 60 to 84% HRR represents vigorous/hard effort, and >84% HRR represents very hard effort. The MVPA threshold of 40% HRR is equivalent to the American College of Sports Medicine's definition of moderate exercise (64 to 76% of *maximal heart rate*) (55).

Statistical analysis. Analyses were performed in R (v4.3.1) (56). Generalized additive mixed models (GAMMs) were fit to allow nonlinear between- and within-person associations of sleep with outcomes. Outcomes were next-day MVPA duration (minutes) and overall PA duration (minutes). Covariates included in all models were weekday (Monday–Friday) and weekend (Saturday–Sunday), sex, and smoothing terms for age, Body Mass Index (BMI), the prior day's outcome variable (to control for autocorrelation and allow for a rigorous test of directionality), and the WHOOP Recovery Score. The WHOOP Recovery Score is a proprietary and algorithmically derived metric (from 0 to 100%) accessible to the user on the smartphone app and is calculated from several inputs, including heart rate, heart rate variability, sleep, and respiratory rate. We adjusted for WHOOP Recovery Score as individuals may use this information to make PA decisions. A smoothing term on the participant-centered WHOOP Recovery Score was included in the model. A random intercept by participant was included in all models to account for nonindependence.

Primary analyses focused on the main effects of sleep on outcomes. The predictors of these models included four smooth functions of the between- and within-person effects of sleep duration and sleep onset timing. All four smooth terms were included in the same model to account for their interrelationships (*SI Appendix*, Tables S1 and S2). The smoothing estimates for between- and within-person sleep were plotted to interpret each GAMM (Figs. 1 and 2). Custom contrasts were built to compare estimated marginal means, where different levels of each predictor were compared to a reference level represented by the nearest interpretable value that approximated the sample mean (e.g., reference levels for sleep duration: between-person = 7 h; within-person = 0 h; reference levels for sleep onset timing: between-person = 11:00 PM; within-person = 0 h). We also built contrasts that compared levels of between-person predictors that were ±2 h of the reference level to examine differences between extremes of our sample (e.g., compared average sleep onset timing of 9:00 PM to 1:00 AM). The *t*-distribution adjustment method was used to adjust for multiple tests. Contrast estimates reflect both relative and absolute differences between two levels of a predictor and therefore provide an interpretable effect size and real-world significance beyond statistical significance.

Secondary analyses tested whether sleep duration and sleep onset timing interacted to predict next-day PA outcomes. Tensor product smooths (57) were used to model GAMM interactions. Predicted next-day MVPA and overall PA durations were plotted on two-dimensional heatmaps featuring within-person sleep onset timing on the x-axis and within-person sleep duration on the y-axis

(Fig. 3). Estimates were plotted at coordinates using 15-min intervals of within-person sleep onset timing and duration from −4 h to +4 h relative to their monthly average.

External Validation Study (All of Us) Methods.

Design, population, and data collection. The All of Us Research Program is a large-scale longitudinal cohort study led by the NIH and designed to collect multimodal health data from participants across the United States, with a key goal of enhancing generalizability of biomedical research through representation from historically underrepresented populations, further described elsewhere (44). Participants provided informed consent, completed demographic, lifestyle, and health surveys, and have linked electronic health records. Participants who previously owned Google Fitbit devices (Googleplex, Mountain View, CA) were invited to share wearable data, and the Wearables Enhancing All of Us Research Study was subsequently launched to provide Fitbit devices at no cost to eligible participants who identified with one or more underrepresented communities in biomedical research (58).

Among 40,776 actively consented All of Us participants with Fitbit data in Controlled Tier Dataset V8 (C2024Q3R4), containing data through October 1, 2023, 5,898 met inclusion criteria, including being aged ≥18 y, having no missing information for sex, age, race, and ethnicity, having logged ≥100 main sleep episodes across a 120-d interval (i.e., ≥83% usage) during the All of Us study, and having <20% of sleep onset and offset times after 12:00 PM local time on the same calendar day. The last criterion was introduced to minimize misclassification of pre- and postsleep PA because PA data were available only as daily summaries, precluding intraday sequential linking. The final dataset included 635,477 person-nights of data (see CONSORT flow diagram, *SI Appendix*, Fig. S1*B*). Participant characteristics are presented in Table 2.

Participant-level daily summary sleep data were measured using Fitbit devices. Performance evaluation of a recent Fitbit device (Fitbit Charge 2™) compared with polysomnography demonstrated that Fitbit overestimated polysomnography total sleep time by 9 min and showed improved performance for accurately detecting wake compared to standard actigraphy-based wearables (59). Activity data were also measured using Fitbit devices, including light, moderate, and vigorous PA and steps. Evaluation of Fitbit devices on individuals in free-living conditions has demonstrated substantially higher measures of PA (i.e., median increase of 52 to 390% versus comparator accelerometers) and slightly higher step counts, with 85% of studies finding Fitbit steps within 10% of measures by comparator accelerometers or pedometers (60).

Covariates. Participant demographics, lifestyle factors, and self-rated health were derived from survey data completed at the time of enrollment. Demographics included categorical variables sex at birth (female or male), combined race and ethnicity (non-Hispanic ethnicity and White race, non-Hispanic Black, non-Hispanic Asian, non-Hispanic other or multiple, and Hispanic ethnicity, any race or races–collapsed to White or non-White for models), and employment (employed, unemployed, or retired–collapsed to employed or not employed for models) and continuous variables annual income (nine levels of increasing income brackets), Deprivation Index (61) (a measure of socioeconomic disadvantage at a community level based on American Community Survey data), cigarette smoking frequency (<100 lifetime cigarettes, ≥100 and currently not every day, ≥100 and currently everyday), alcohol use frequency (never, weekly or less, multiple drinks weekly), and self-rated general health (poor, fair, good, very good, or excellent). BMI ascertained from electronic health records as the BMI recorded closest to the study interval start date. For GAMMs, missing covariate data were imputed with multiple decision tree-based imputations (n = 5) using the CART method in the *mice* R package, with postprocessing constraints to variable upper and lower bounds.

Predictors: Sleep duration and sleep onset timing. Fitbit devices use accelerometry and heart rate variability in a proprietary interpretative algorithm for sleep/wake classification. Sleep duration was operationalized as the total minutes asleep during daily summary sleep data. Sleep onset timing was operationalized as the first occurrence of a state corresponding to sleep (i.e., asleep, light, deep, rapid eye movement) during the main sleep episode for each day, determined by the Fitbit algorithm based on the longest sleep per day. Between- and within-person terms were consistent with the primary study.

Outcomes: MVPA, overall PA, and step count. Outcomes were next-day metrics derived from daily summaries of total MVPA duration, overall (i.e., light, moderate, vigorous) PA duration, and step count.

Statistical analysis. Analyses were performed in R (v.4.2.2) on the All of Us Researcher Workbench, a secure cloud-based platform. As in the primary analysis, GAMMs were fit to allow nonlinear between- and within-person associations of sleep duration and onset with each outcome: next-day MVPA duration, overall PA duration, and step count. GAMMs included a random intercept by participant and the following covariates: weekday (Monday–Friday) and weekend (Saturday–Sunday), sex, combined race and ethnicity, and employment, and smoothed continuous age, income, Deprivation Index, cigarette smoking frequency, alcohol use frequency, BMI, self-rated general health, and prior day's outcome variable. Smoothing estimates for between- and within-person sleep were plotted to interpret each GAMM, and custom contrasts were built to compare estimated marginal means.

Data, Materials, and Software Availability. The dataset associated with the primary study comprises regularly collected physiological and healthcare data as well as demographic information and is stored in a proprietary repository. Given its sensitive nature and the potential for reidentification, access to the dataset is enforced through an application process. For inquiries regarding data access, please contact support@whoop.com. All data requests will be assessed and addressed in accordance with policies designed to safeguard participant and user confidentiality, as outlined in the terms and conditions and informed consent documentation. The timeframe for response to requests will be 4 wk. The dataset associated with the external validation study comprises regularly collected physiological and healthcare data as well as demographic information and can be accessed via application of a Data Use and Registration Agreement with the All of Us Research Program by emailing aoudurasupport@vumc.org.

ACKNOWLEDGMENTS. The authors did not receive specific funding for this work. J.L. and E.R.F.-C. received support from the Wu Tsai Human Performance Alliance and the Joe and Clara Tsai Foundation. J.L. received financial support from the Australian Government through Research Training Program scholarships. F.L. receives financial support from the Monash Graduate Scholarship and the Monash International Tuition Scholarship. M.É.C. received support from the Australian-American Fulbright Foundation, with funding provided by The Kinghorn Foundation. H.S. reports research support *unrelated to this work* from Re-Time Pty Ltd., Withings Ltd., Compumedics Ltd., the American Academy of Sleep Medicine Foundation, and Flinders University. J.F.W. was supported by the National Health and Medical Research Council (NHMRC) fellowship (#1178487). E.R.F-C is financially supported by an Australian Research Council Industry Fellowship (IE240100162) and has received research support or consultancy fees from the Science Industry Endowment Fund, Monash Lung and Sleep Institute, Tempur Australia, Team Focus Ltd., British Athletics, Australian National Football League, Australian National Rugby League, Collingwood Football Club, Melbourne Storm Rugby Club, and Henley Business School, which are not related to this paper. S.M.W.R. has served as a Programme Leader for the Cooperative Research Centre for Alertness, Safety and Productivity, Australia; is a Director and Chair of the Sleep Health Foundation; has received grants from Vanda Pharmaceuticals, Philips Respironics, Cephalon, Rio Tinto, BHP Billiton, and Shell; and has received equipment support and consultancy fees through his institution from Optalert, Compumedics, Teva Pharmaceuticals, Roche, and Circadian Therapeutics, which *are not related* to this paper. S.P.A.D. has received funding from the NHMRC and the United States Department of Defence, received consultancy fees from Avecho Biotechnology Ltd., and participated on an advisory board for Zelda Therapeutics, which *are not related* to this paper.

Author affiliations: [a]Sleep and Circadian Rhythms Research Program, School of Psychological Sciences, Faculty of Medicine, Nursing and Health Sciences, Monash University, Melbourne, VIC 3800, Australia; [b]Division of Sleep and Circadian Disorders, Department of Medicine, Brigham and Women's Hospital, Boston, MA 02115; [c]Francis Weld Peabody Society, Harvard Medical School, Boston, MA 02115; [d]Department of Data Science and Research, Whoop, Inc., Boston, MA 02215; [e]Swiss Institute of Bioinformatics, Lausanne, 1015, Switzerland; [f]Department of Computational Biology, University of Lausanne, Lausanne 1015, Switzerland; and [g]Flinders Health and Medical Research Institute: Sleep Health, College of Medicine and Public Health, Flinders University, Adelaide, SA 5042, Australia

Author contributions: J.L. and M.É.C. designed research; J.L., M.É.C., D.M.P., E.R.C., performed research; D.M.P. contributed new reagents/analytic tools; J.L., M.É.C., F.L., and J.F.W. analyzed data; M.É.C., F.L., D.M.P., E.R.C., H.S., J.F.W., S.P.A.D., S.M.W.R., and E.R.F.-C. edited the paper; and J.L. and M.É.C. wrote the paper.

1. P. M. Kris-Etherton *et al.*, Strategies for promotion of a healthy lifestyle in clinical settings: Pillars of ideal cardiovascular health: A science advisory from the American Heart Association. *Circulation (New York, N.Y.)* **144**, E495–E514 (2021).
2. G. E. Vincent *et al.*, Improving cardiometabolic health with diet, physical activity, and breaking up sitting: What about sleep? *Front. Physiol.* **8**, 865–865 (2017).
3. M. M. Badran, B. A. Yassin, N. Fox, I. Laher, N. Ayas, Epidemiology of sleep disturbances and cardiovascular consequences. *Can. J. Cardiol.* **31**, 873–879 (2015).
4. A. Kankanhalli, M. Saxena, B. Wadhwa, Combined interventions for physical activity, sleep, and diet using smartphone apps: A scoping literature review. *Int. J. Med. Inform.* **123**, 54–67 (2019).
5. L. K. Barger, J. K. P. Wright, R. J. Hughes, C. A. Czeisler, Daily exercise facilitates phase delays of circadian melatonin rhythm in very dim light. *Am. J. Physiol. Integr. Comp. Physiol.* **286**, R1077–R1084 (2004).
6. M. Chennaoui, P. J. Arnal, F. Sauvet, D. Léger, Sleep and exercise: A reciprocal issue? *Sleep Med. Rev.* **20**, 59–72 (2015).
7. Y. Yang, J. C. Shin, D. Li, R. An, Sedentary behavior and sleep problems: A systematic review and meta-analysis. *Int. J. Behav. Med.* **24**, 481–492 (2017).
8. J. Craven *et al.*, Effects of acute sleep loss on physical performance: A systematic and meta-analytical review. *Sports Med. (Auckland)* **52**, 2669–2690 (2022).
9. J. Charest, M. A. Grandner, Sleep and athletic performance: Impacts on physical performance, mental performance, injury risk and recovery, and mental health: An update. *Sleep Med. Clin.* **17**, 263–282 (2022).
10. S. L. Halson, Nutrition, sleep and recovery. *Eur. J. Sport Sci.* **8**, 119–126 (2008).
11. Y. Liu *et al.*, Prevalence of healthy sleep duration among adults–United States, 2014. *MMWR. Morb. Mortal. Wkly. Rep.* **65**, 137–141 (2016).
12. K. L. Piercy *et al.*, The physical activity guidelines for Americans. *JAMA* **320**, 2020–2028 (2018).
13. S. Atoui *et al.*, Daily associations between sleep and physical activity: A systematic review and meta-analysis. *Sleep Med. Rev.* **57**, 101426 (2021).
14. L. E. Bromley, J. N. Booth, J. M. Kilkus, J. G. Imperial, P. D. Penev, Sleep restriction decreases the physical activity of adults at risk for type 2 diabetes. *Sleep* **35**, 977–984
15. S. M. Schmid *et al.*, Short-term sleep loss decreases physical activity under free-living conditions but does not increase food intake under time-deprived laboratory conditions in healthy men. *Am. J. Clin. Nutr.* **90**, 1476–1482 (2009).
16. P. Bernard, H. Ivers, M.-H. Savard, J. Savard, Temporal relationships between sleep and physical activity among breast cancer patients with insomnia. *Health Psychol.* **35**, 1307–1315 (2016).
17. M. P. Mead, K. Baron, M. Sorby, L. A. Irish, Daily associations between sleep and physical activity. *Int. J. Behav. Med.* **26**, 562–568 (2019).
18. M. M. S. Kishida, S. P. Elavsky, An intensive longitudinal examination of daily physical activity and sleep in midlife women. *Sleep Health* **2**, 42–48 (2016).
19. J. Fanning *et al.*, Physical activity, mind wandering, affect, and sleep: An ecological momentary assessment. *JMIR Mhealth Uhealth* **4**, e104 (2016).
20. J. A. Mitchell *et al.*, No evidence of reciprocal associations between daily sleep and physical activity. *Med. Sci. Sports Exerc.* **48**, 1950–1956 (2016).
21. J. R. Best, R. S. Falck, G. J. Landry, T. Liu-Ambrose, Analysis of dynamic, bidirectional associations in older adult physical activity and sleep quality. *J. Sleep Res.* **28**, e12769 (2019).
22. L. A. Irish *et al.*, A 24-hour approach to the study of health behaviors: Temporal relationships between waking health behaviors and sleep. *Ann. Behav. Med.* **47**, 189–197 (2014).
23. K. G. Baron, K. J. Reid, Circadian misalignment and health. *Int. Rev. Psychiatry* **26**, 139–154 (2014).
24. C. L. Guarana, J. W. Ryu, E. H. O'Boyle, J. Lee, C. M. Barnes, Sleep and self-control: A systematic review and meta-analysis. *Sleep Med. Rev.* **59**, 101514 (2021).
25. S. L. Chellappa, Circadian misalignment: A biological basis for mood vulnerability in shift work. *Eur. J. Neurosci.* **52**, 3846–3850 (2020).
26. I. Janssen *et al.*, A systematic review of compositional data analysis studies examining associations between sleep, sedentary behaviour, and physical activity with health outcomes in adults. *Appl. Physiol. Nutr. Metab.* **45**, S248–S257 (2020).
27. P. J. Curran, D. J. Bauer, The disaggregation of within-person and between-person effects in longitudinal models of change. *Annu. Rev. Psychol.* **62**, 583–619 (2011).
28. K. R. Evenson, F. Wen, J. S. Metzger, A. H. Herring, Physical activity and sedentary behavior patterns using accelerometry from a national sample of United States adults. *Int. J. Behav. Nutr. Phys. Act.* **12**, 20 (2015).
29. S. M. Mattingly *et al.*, The effects of seasons and weather on sleep patterns measured through longitudinal multimodal sensing. *NPJ Digital Med.* **4**, 76 (2021).
30. Y. Ma *et al.*, Association between sleep duration and cognitive decline. *JAMA Netw. Open* **3**, e2013573 (2020).
31. C. Krittanawong *et al.*, Association between short and long sleep durations and cardiovascular outcomes: A systematic review and meta-analysis. *Eur. Heart J. Acute Cardiovasc. Care* **8**, 762–770 (2019).
32. A. Garriga, N. Sempere-Rubio, M. J. Molina-Prados, R. Faubel, Impact of seasonality on physical activity: A systematic review. *Int. J. Environ. Res. Public Health* **19**, 2 (2021).
33. A.-K. Pesonen *et al.*, Sleep and physical activity–The dynamics of bi-directional influences over a fortnight. *BMC Public Health* **22**, 1160 (2022).
34. E. G. Heiland *et al.*, Bi-directional, day-to-day associations between objectively-measured physical activity, sedentary behavior, and sleep among office workers. *Int. J. Environ. Res. Public Health* **18**, 7999 (2021).
35. S. D. Youngstedt, C. E. Kline, Epidemiology of exercise and sleep. *Sleep Biol. Rhythms* **4**, 215–221 (2006).
36. F. Le, Y. Yap, N. Y. C. Tung, B. Bei, J. F. Wiley, The associations between daily activities and affect: A compositional isotemporal substitution analysis. *Int. J. Behav. Med.* **29**, 456–468 (2022).
37. N. Sempere-Rubio, M. Aguas, R. Faubel, Association between chronotype, physical activity and sedentary behaviour: A systematic review. *Int. J. Environ. Res. Public Health* **19**, 9646 (2022).
38. I. Merikanto, L. Kuula, J. Lahti, K. Räikkönen, A.-K. Pesonen, Eveningness associates with lower physical activity from pre- to late adolescence. *Sleep Med.* **74**, 189–198 (2020).
39. F. Giannotti, F. Cortesi, T. Sebastiani, S. Ottaviano, Circadian preference, sleep and daytime behaviour in adolescence. *J. Sleep Res.* **11**, 191–199 (2002).
40. M. Wittmann, J. Dinich, M. Merrow, T. Roenneberg, Social jetlag: Misalignment of biological and social time. *Chronobiol. Int.* **23**, 497–509 (2006).
41. J. Leota *et al.*, Dose-response relationship between evening exercise and sleep. *Nat Commun.* **16**, 3297 (2025).
42. E. R. Facer-Childs, B. Middleton, D. J. Skene, A. P. Bagshaw, Resetting the late timing of 'night owls' has a positive impact on mental health and performance. *Sleep Med.* **60**, 236–247 (2019).
43. A. Maloney, S. Mengesteab, N. Kallas, M. Bennett, J. A. Kanaley, Sleep restriction by sleep timing late night or early wake: The impact on physical activity and dietary intake in adults. *Appetite* **189**, 106996 (2023).
44. J. C. Denny *et al.*, The "All of Us" research program. *N. Engl. J. Med.* **381**, 668–676 (2019).
45. S. Skender *et al.*, Accelerometry and physical activity questionnaires–A systematic review. *BMC Public Health* **16**, 515 (2016).
46. S. A. Adams *et al.*, The effect of social desirability and social approval on self-reports of physical activity. *Am. J. Epidemiol.* **161**, 389–398 (2005).
47. H. Feng *et al.*, Associations of timing of physical activity with all-cause and cause-specific mortality in a prospective cohort study. *Nat. Commun.* **14**, 930 (2023).
48. P. H. Lee, A. C. Y. Tse, C. S. T. Wu, Y. W. Mak, U. Lee, Temporal association between objectively measured smartphone usage, sleep quality and physical activity among Chinese adolescents and young adults. *J. Sleep Res.* **30**, e13213 (2021).
49. S. A. A. Massar *et al.*, Reopening after lockdown: The influence of working-from-home and digital device use on sleep, physical activity, and wellbeing following COVID-19 lockdown and reopening. *Sleep* **45**, zsab250 (2022).
50. D. P. Windred *et al.*, Sleep regularity is a stronger predictor of mortality risk than sleep duration: A prospective cohort study. *Sleep* **47**, zsad253 (2024).
51. D. J. Miller, C. Sargent, G. D. Roach, A validation of six wearable devices for estimating sleep, heart rate and heart rate variability in healthy adults. *Sensors (Basel)* **22**, 6317 (2022).
52. A.-M. Schyvens *et al.*, Accuracy of fitbit charge 4, Garmin Vivosmart 4, and WHOOP Versus polysomnography: Systematic review. *JMIR Mhealth Uhealth* **12**, e52192 (2024).
53. D. J. Miller *et al.*, A validation study of a commercial wearable device to automatically detect and estimate sleep. *Biosensors (Basel)* **11**, 185 (2021).
54. D. E. R. Warburton, P. T. Katzmarzyk, R. E. Rhodes, R. J. Shephard, Evidence-informed physical activity guidelines for Canadian adults. *Appl. Physiol. Nutr. Metab.* **98**, S16–S68 (2007).
55. G. Liguori, A. C. o. S. Medicine, *ACSM's Guidelines for Exercise Testing and Prescription* (Lippincott williams & wilkins, 2020).
56. R. C. Team, RA language and environment for statistical computing (Version v4.3.1, R Foundation for Statistical Computing Vienna, Austria, 2020).
57. S. N. Wood, Low-rank scale-invariant tensor product smooths for generalized additive mixed models. *Biometrics* **62**, 1025–1036 (2006).
58. H. Jeong *et al.*, Data from the All of Us research program reinforces existence of activity inequality. *NPJ Digital Med.* **8**, 8 (2025).
59. M. de Zambotti, A. Goldstone, S. Claudatos, I. M. Colrain, F. C. Baker, A validation study of Fitbit Charge 2™ compared with polysomnography in adults. *Chronobiol. Int.* **35**, 465–476 (2018).
60. L. M. Feehan *et al.*, Accuracy of fitbit devices: Systematic review and narrative syntheses of quantitative data. *JMIR Mhealth Uhealth* **6**, e10527 (2018).
61. C. Brokamp *et al.*, Material community deprivation and hospital utilization during the first year of life: An urban population–based cohort study. *Ann. Epidemiol.* **30**, 37–43 (2019).